

AI-Based Manufacturing Efficiency Classification

Using Sensor, Production & 6G Network Data

Executive Summary Report

Generated: April 30, 2026 at 01:16 AM

Thales Group | Smart Manufacturing Division

1. Dataset Overview

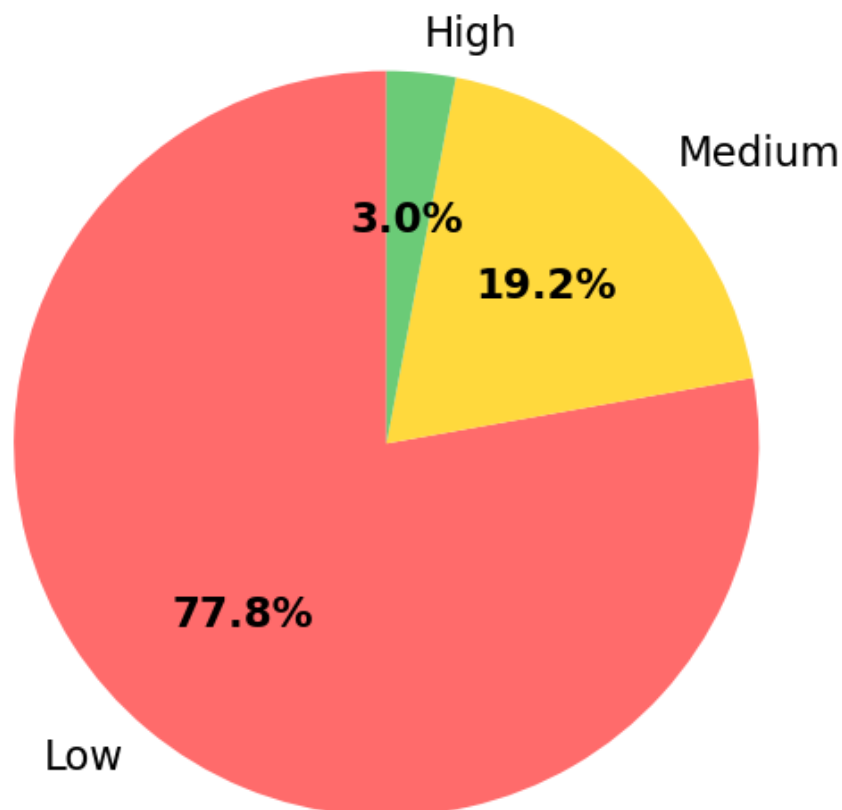
The analysis is based on the Thales Group Manufacturing dataset containing 100,000 operational records from 50 industrial machines, collected over 69 days with minute-level granularity.

Total Records	Machines	Days	Features
100,000	50	69	14

Class Distribution

The target variable (Efficiency_Status) shows significant class imbalance: Low (77.8%), Medium (19.2%), High (3.0%). SMOTE oversampling was applied to address this imbalance.

Efficiency Status Distribution

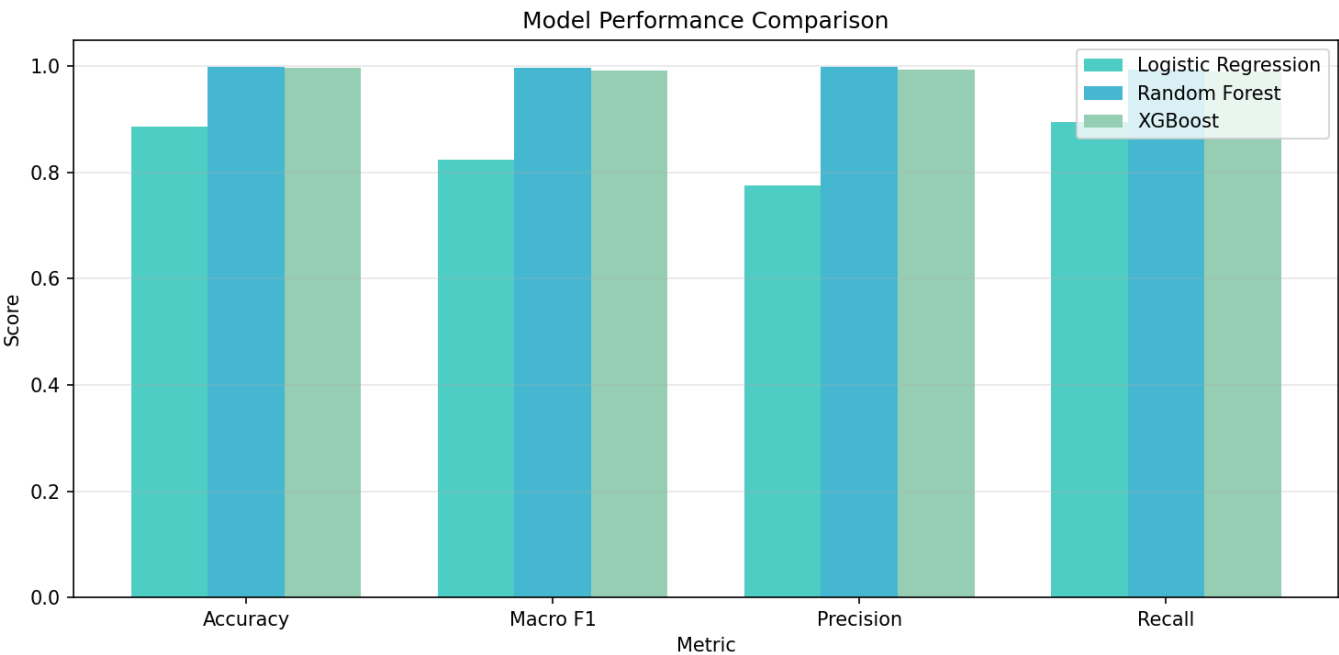


2. Model Performance

Three classification models were trained and evaluated using temporal train/test split (training on earlier data, testing on later data) with macro-averaged F1 as the primary metric.

Model	Accuracy	Macro F1	Wt F1	Precision	Recall	CV F1
Logistic Regression	0.8868	0.8247	0.8928	0.7760	0.8953	0.9212
Random Forest	0.9994	0.9965	0.9994	0.9990	0.9940	0.9982
XGBoost	0.9969	0.9921	0.9969	0.9946	0.9897	0.9978

Best performing model: Random Forest

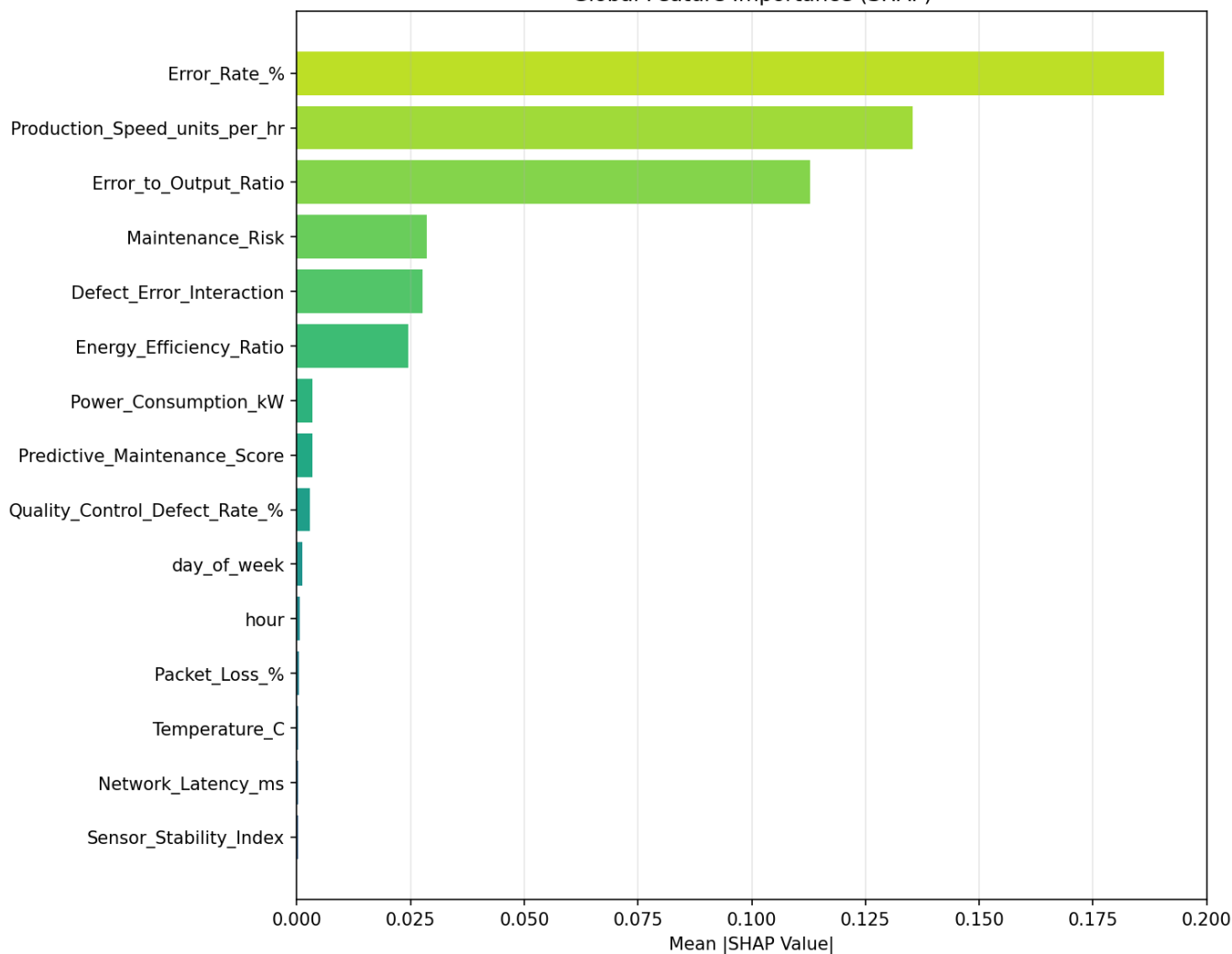


3. Feature Importance

SHAP (SHapley Additive exPlanations) analysis reveals the most influential features driving efficiency classification decisions:

#	Feature	Importance (Mean SHAP)
1	Error_Rate_%	0.190572
2	Production_Speed_units_per_hr	0.135403
3	Error_to_Output_Ratio	0.112834
4	Maintenance_Risk	0.028616
5	Defect_Error_Interaction	0.027613
6	Energy_Efficiency_Ratio	0.024506
7	Power_Consumption_kW	0.003502
8	Predictive_Maintenance_Score	0.003374
9	Quality_Control_Defect_Rate_%	0.002799
10	day_of_week	0.001285

Global Feature Importance (SHAP)



4. Key Findings & Recommendations

1. Error_Rate_% is the most influential factor determining manufacturing efficiency, followed by Production_Speed_units_per_hr and Error_to_Output_Ratio.
2. The Random Forest model achieves the best balance of precision and recall across all efficiency classes, making it suitable for production deployment.
3. Severe class imbalance (High efficiency: ~3%) was successfully addressed using SMOTE oversampling, enabling the model to identify rare high-efficiency states.
4. Temporal validation confirms the model generalizes to future data, supporting real-time deployment in factory control systems.
5. Network metrics (latency, packet loss) contribute to classification, validating the importance of 6G connectivity in smart manufacturing.

Recommendations

- Deploy the trained model as a real-time efficiency monitoring service.
- Set up automated alerts when efficiency drops below Medium threshold.
- Focus maintenance efforts on machines with high Error_Rate and low Maintenance Scores.
- Monitor network reliability as it directly impacts classification accuracy.
- Retrain the model quarterly with new data to maintain prediction accuracy.
- Investigate root causes behind the dominant Low efficiency status (77.8%).